



CONTACT

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Portfolio: rtlcoding.com

Verified Reviews:

[upwork.com/freelancers/rowanlumb](https://www.upwork.com/freelancers/rowanlumb)

Schedule a Call: cal.id/rowan-thomas-edward-lumb/upwork-east

SKILLS

Programming

R, Python, Angular/Typescript, SQL, C++, Google Cloud Platform

Design

Computational Modeling (FEA), AutoDesk Fusion, OpenCV (computer vision)

Mathematics

Calculus, Differential Equations (DEs/PDEs), Linear Algebra, Statistics/Probability

Rowan Thomas Lumb

US & UK DUAL CITIZEN

AUTOMATION & DATA ENGINEER | M.SC. MECHANICAL ENGINEERING

Engineer/Developer with a background spanning software development, mechanical engineering, computational modeling, CAD, data analysis, and experimental research. Interested in actively moving into the interface of software and hardware and physical product development.

WORK EXPERIENCE

Engineer/Developer

[RTL Software LLC \(Self-Employed\)](#)

August 2019 — Present

Extremely varied project-based work ranging from simple workflow automation and mobile app data processing to full-stack SaaS implementations. Software development, automation, data analysis, and engineering-oriented computational work.

Software Developer

[ViewGlass](#)

May 2018 — August 2018

Developed full-stack web applications for manufacturing-line operations in a .NET environment, utilizing RESTful APIs and Angular 2/Bootstrap. Applications were deployed in a manufacturing environment.

Graduate Teaching Assistant

[University of Memphis](#)

June 2017 — May 2018

Led engineering labs while working toward a Master's Degree in Mechanical Engineering. Taught differential equations in mechanical, thermal, and electronic applications. Performed research using Big Data and artificial neural networks.

Research Assistant

[Los Alamos National Laboratory](#)

June 2016 — August 2016

Assisted in magnetic mapping of UCNtau, an experiment to precisely measure the half-life (tau) of the free neutron apparatus, to analyze possible systematic effects.

Research Assistant

[Los Alamos National Laboratory](#)

June 2015 — August 2015

Developed precise 3-D object tracking software using C++ and OpenCV to monitor object location within an experimental apparatus.

Research Assistant

[Oak Ridge National Laboratory](#)

June 2014 — August 2014

Assisted in construction of a Fast Ionization Chamber for new Gammasphere and ORRUBA nuclear detectors. Developed predictive R-process simulations using ROOT.

EDUCATION

M.Sc. in Mechanical Engineering

[University of Memphis](#)

June 2017 — May 2019

Research/thesis work in acoustic emission non-destructive testing of fatigued 4340 steel and 7075 aluminum. Data analysis utilized R and supervised neural networks. Research published in Springer's Data Enabled Discovery journal: <https://link.springer.com/article/10.1007/s41688-020-0036-7>

B.Sc. in Physics

[Tennessee Technological University](#)

August 2012 — May 2017

References available upon request.